

A PLAIN-LANGUAGE GUIDE

Understanding Dose & Risk

How researchers work out whether "more" really means "worse" — and why the method they choose changes the answer.

Based on: Heiko Becher & Matthias Schmid, "Analysis of Continuous Covariates and Dose-Effect Analysis," Handbook of Epidemiology (Springer, 2024)

WHY THIS TOPIC MATTERS

Most health risks come in amounts, not yes-or-no answers

Grams of alcohol per day. Cigarettes smoked. Blood pressure. Body weight. None of these are simple on/off switches — they are measured on a scale, and risk can climb gradually, level off, or jump at certain points along that scale.

How a researcher chooses to handle those numbers — not just what the data says — can change the conclusion a study reaches about whether something is dangerous, and how dangerous it is.

This deck's promise

Every method a health researcher actually uses to study "how much" — explained without formulas, using two real cancer studies as running examples.

THE EVIDENCE BEHIND THIS DECK

The studies behind every example

Every method in this deck is illustrated with real data from case-control studies of laryngeal (throat and voice-box) cancer — studies that compare people who developed the cancer ("cases") with similar cancer-free people ("controls"), then look back at what they were exposed to.



Germany, 1990s

236 laryngeal-cancer patients and 702 cancer-free controls. Used to study daily alcohol intake and cumulative smoking (in pack-years).



Poland, 1991

249 laryngeal-cancer patients and 965 controls. Used to show how carefully adjusting for age changes what we conclude about smoking.

SETTING THE STAGE

How researchers approach the analysis

STEP ONE

Look before you test

A careful, descriptive look at the data — simple summaries and plots — almost always comes first, and can save enormous amounts of time later.

STEP TWO

Then model, carefully

Formal statistical modeling follows. The chapter warns against rushing to a "significant" result, or trying many models until one happens to look good.

There are also two different reasons to analyze a continuous measurement: because it is the very thing you want to understand (does smoking raise cancer risk, and by how much?), or because it is a factor you need to correctly account for so it does not distort your answer about something else.

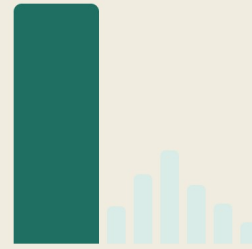
A RECURRING WRINKLE

The “zero” problem

For exposures like smoking or drinking, a large share of people have exactly none — then everyone else spreads across a continuous range. Statisticians call this a **spike at zero**.

This matters because familiar tricks — like taking a logarithm of the exposure — break down at exactly zero, and the jump from “none” to “a little” may not behave the same way as the jump from “a little” to “a lot.” We will come back to this with a purpose-built solution later.

EXAMPLE: A SMOKING STUDY



One tall bar of nonsmokers, then a spread of smoking amounts among everyone else (illustrative histogram, author-created)

CLASSIC METHOD

Grouping Into Categories

For decades, the standard approach was to sort everyone into a handful of groups — say, none, low, medium, and high — and compare disease rates across the groups against a chosen baseline.

OPEN QUESTION 1

How many groups?

OPEN QUESTION 2

Where do the lines go?

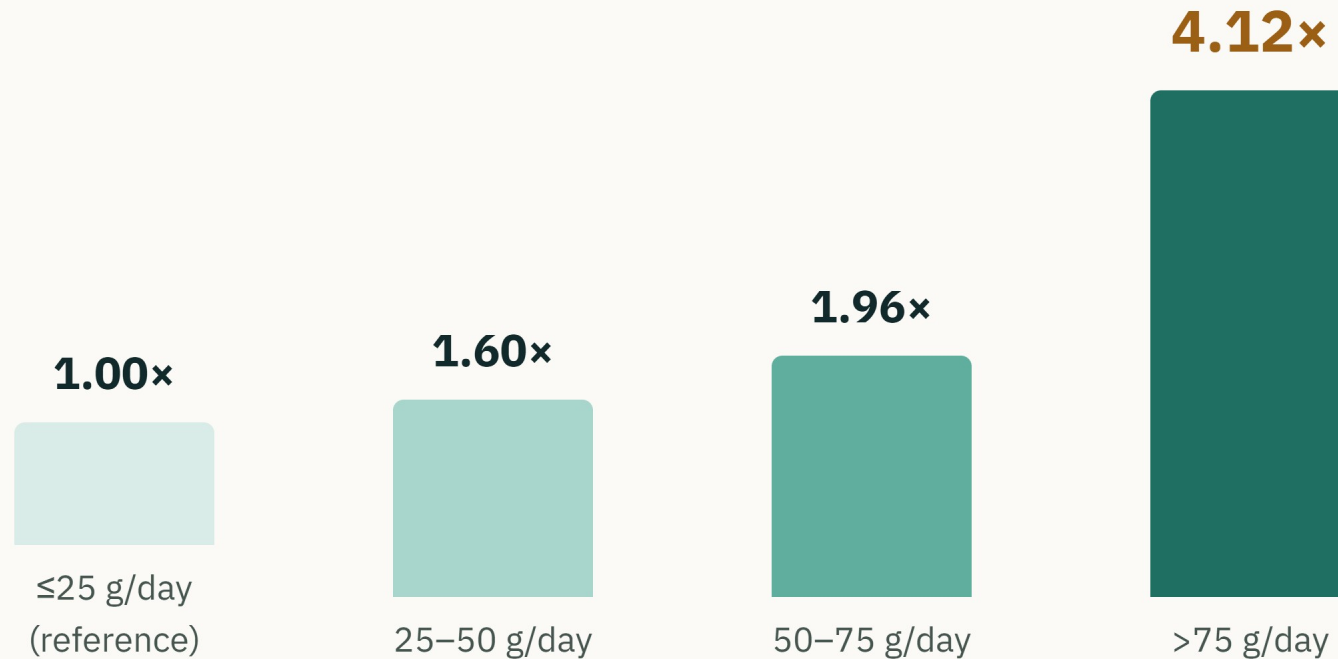
RULES OF THUMB FROM THE CHAPTER

- Match cutoffs used in earlier studies, so results are comparable
- Use a natural baseline when one exists (e.g., nonsmokers, non-drinkers)
- Three to six groups usually works well; rarely more than ten
- Decide the groups before looking at results — never after

REAL EXAMPLE · GERMAN CASE-CONTROL STUDY

Alcohol and Throat Cancer, by Group

236 throat-cancer patients and 702 cancer-free controls were grouped by daily alcohol intake. Risk is shown relative to the lightest-drinking group, set at 1.0×



READING THIS CHART

The heaviest drinkers (more than 75 grams of ethanol — roughly six or more standard drinks — per day) had about four times the throat-cancer risk of the lightest drinkers, with a clear step up at every level.

“Risk” here is an odds ratio — how many times higher the odds of cancer are versus the reference group. For the heaviest drinkers, the true value is estimated to fall between 2.77× and 6.12× with 95% confidence. Chart is an author-created illustration of the data in Table 2, not a reproduction of the original figure.

IS THE PATTERN REAL?

Confirming the Pattern Is Real

A step-up across groups could be a coincidence of this one sample. Statisticians use formal “trend tests” to check whether an increase is genuine and consistent — not random noise.

TEST 1 • MANTEL-HAENSZEL

$p < 0.001$

Highly significant

TEST 2 • COCHRAN-ARMITAGE

$p < 0.001$

Highly significant — confirms test 1

In plain terms: a pattern this strong would appear by pure chance less than once in a thousand times if alcohol and throat-cancer risk were truly unrelated. Both tests agree — this dose-response pattern is statistically well supported. Even so, a significant test doesn’t prove risk rises smoothly throughout — it can stay significant even if risk climbs early, then flattens or dips.

THE TRADE-OFF

Where Grouping Falls Short



Information gets lost

Two very different amounts inside the same group are treated as identical.



Cutoffs are arbitrary

Different researchers can draw different lines and get different-looking results from the same data.



Hard to juggle several factors

Adjusting for more than one or two other risk factors at once gets difficult.



Little insight into shape

It shows risk goes up, but reveals almost nothing about the actual shape of that rise.

Still useful: this approach is simple, transparent, and a reasonable first look at the data before moving to the more advanced methods ahead.

A BETTER WAY

Modeling Risk Directly

Instead of chopping measurements into groups, mathematical models can use the exact recorded numbers to estimate a smooth risk curve — while adjusting for other factors like age or sex at the same time.

YES-OR-NO OUTCOMES

Does a person develop the disease?
 (“logistic” models — used throughout this deck)

RATES AND COUNTS

How many cases occur over time in a group? (“Poisson” models)

TIME UNTIL AN EVENT

How long until the disease occurs, if it does? (“survival” models)

Different outcome, same core idea: estimate how risk changes as the exposure changes, using every data point instead of a handful of buckets. The next slides all use the yes-or-no version, since that fits the cancer examples.

SIMPLEST REGRESSION VERSION

Using the Raw Number As-Is

The most basic model enters the exposure exactly as measured, with one number summarizing the whole relationship. It is simple and uses all the information — but it assumes risk changes at one constant percentage rate for every extra unit, all the way along the scale.

ALCOHOL EXAMPLE

+0.9%

higher odds of throat cancer for
every extra gram of ethanol per day

THE CATCH

If that constant-rate assumption is wrong, the resulting curve misrepresents the truth — and can even throw off how well the model adjusts for other factors, like age or sex, at the same time.

A MORE FLEXIBLE MODEL

Transforming the Number First

Sometimes the raw number is reshaped — using its square root, its square, or its logarithm, for example — before it enters the model. The transformation chosen implies a very different-shaped relationship between dose and risk.



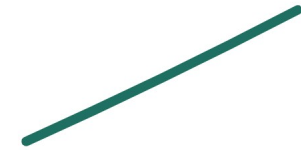
Square root

Rises fast at first, then levels off



Square

Stays low, then shoots up steeply



Logarithm

Rises steadily, close to a straight line

Author-created illustrative shapes, not exact data — the point is how much the choice itself changes the story.

Because the choice of transformation matters this much, researchers need a disciplined way to pick it — not a guess. That is what the next method provides.

A DISCIPLINED WAY TO CHOOSE

Letting Statistics Choose the Best Shape

“Fractional polynomials” are a systematic method: test a defined menu of candidate curve shapes — straight line, square root, log, square, and combinations of two — and use a step-by-step statistical test to decide how complex the curve really needs to be.

1

Is there any effect at all?

Compare the best curve to
“no relationship.”

2

Is a straight line enough?

Compare the best curve to a plain
straight line.

3

Does it need two parts?

Compare a one-part curve to a more
complex two-part curve.

This avoids eyeballing or guessing the right shape — the data itself, through formal tests, decides how simple or complex the final curve should be. A one-part (FP1) curve is always monotonic; a two-part (FP2) curve can rise and then fall.

Smoking Dose and Throat Cancer Risk

Applying the three-step test to smoking amount (in pack-years) among male smokers:

STEP 1

Real effect? Yes

$p < 0.001$

STEP 2

Straight line enough? No

$p < 0.001$

STEP 3

Needs two parts? No

$p = 0.084$ (not significant)



Best-fitting shape: risk rises with the **logarithm** of pack-years smoked — increasing quickly at first, then more gradually — not as a plain straight line.

BACK TO THE ZERO PROBLEM

Handling the Zero Group Properly

For exposures with a spike at zero — like smoking, where many people are nonsmokers — researchers combine two pieces, then formally test which parts are actually needed:

PIECE A

An “exposed at all” flag

A simple yes/no marker for whether someone was exposed at all.

PIECE B

A dose curve among the exposed

The best-fitting curve, fit only among the people with some exposure.

TWO-STAGE TEST

- Stage 1: Do the flag and the curve, together, explain anything at all?
- Stage 2: Once both are in, is each piece still pulling its weight — or can one be dropped?

REAL EXAMPLE • NONSMOKERS INCLUDED

Smoking Risk, Nonsmokers Included

Applying the two-piece method to the full group — smokers and nonsmokers together — and then testing whether each piece still earns its place:

CAN WE DROP THE “ANY SMOKING” FLAG?

Yes, safely

$p = 0.433$ — not significant on its own

CAN WE DROP THE DOSE CURVE?

No — essential

$p < 0.001$ — highly significant

- ✓ The amount someone smoked (on a log scale) explains the risk on its own. Simply knowing “smoker vs. nonsmoker” adds nothing further once dose is in the picture.

A DIFFERENT GOAL

When the Variable Is Just a Nuisance

Sometimes a continuous variable — like age — is not what you are studying. It is a **confounder**: something that must be controlled for so it does not distort the result for the risk factor you actually care about, such as smoking.

Every method covered so far can adjust for a confounder, but none can fully rule out leftover distortion. A more thorough approach — **conditioning** — compares each case and control only against very closely matched peers, so the confounder's influence cancels out mathematically instead of being estimated and subtracted only approximately.



The idea, in one line

Compare a 45-year-old case only to 45-year-old controls, a 62-year-old case only to 62-year-old controls, and so on — then combine those fine-grained comparisons.



The trade-off

Very fine matching groups can end up with too few people in them, which can waste some statistical power.

Age Adjustment: Precision Matters

Researchers studying smoking and throat cancer had to account for age. As the age adjustment got more precise, the estimated strength of the smoking effect changed:

Method of adjusting for age	Estimated smoking effect
No adjustment for age at all	1.73
Rough grouping (2 age bands)	1.44
Finer grouping (7 age bands)	1.24
Smooth statistical curve (fractional polynomial)	1.19
Finest possible: matched within a single year of age	1.16

Takeaway: the cruder the age adjustment, the more of age’s own effect leaked into — and inflated — the smoking estimate. The most careful adjustment gave the smallest, most trustworthy answer.

THE MOST FLEXIBLE OPTION

Let the Data Draw the Curve

Rather than picking from a fixed menu of transformations, the most flexible modern approach — using tools called generalized additive models and splines — estimates the shape of the relationship directly from the data, with very few assumptions decided in advance.

Think of it like bending a flexible strip so it naturally follows wherever the data points lead, instead of forcing a straight ruler or a pre-cut template onto them:



Too rigid

A straight line can miss a real bend in the data



Just right

A smooth curve that follows the real pattern



Too wiggly

Overreacts to random noise in the sample

A built-in tuning setting keeps the curve balanced — smooth enough to trust, flexible enough to catch a real bend. (Diagrams above are author-created illustrations.)

WHAT'S ACTUALLY INSIDE THE TUNING SETTING

How a Spline Stays Flexible Without Chasing Noise

A spline builds its curve from short, simple pieces joined at marker points called **knots**. How many knots there are, and how much freedom each piece gets, decides whether the final curve stays trustworthy.

Too few knots

The pieces are too long and rigid to bend where the real pattern turns — the same problem as “too rigid” earlier.

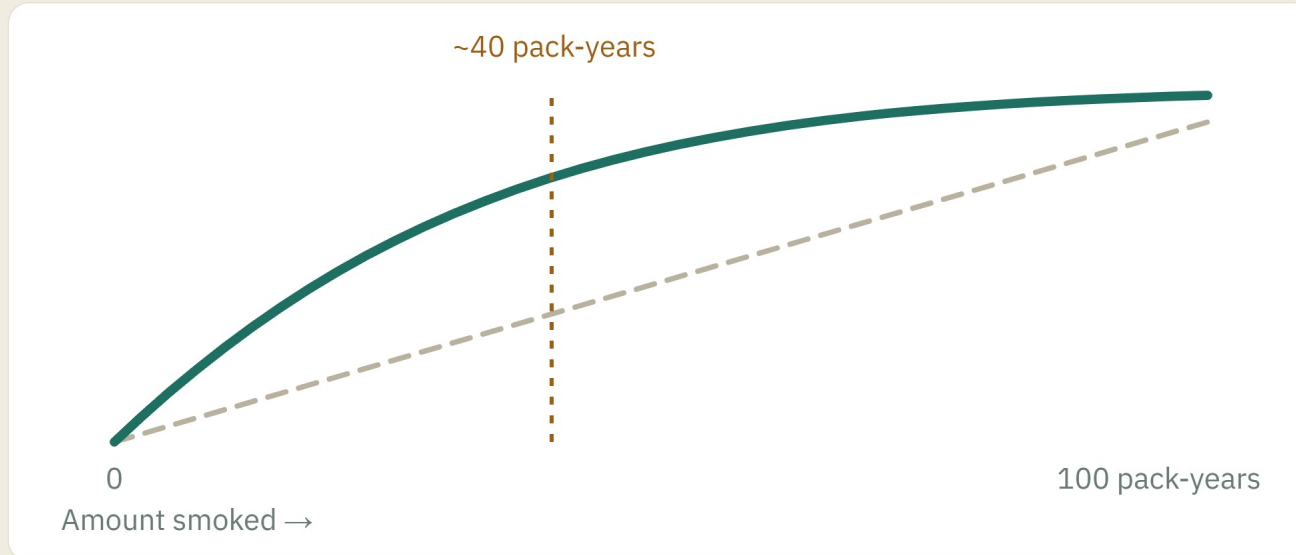
Too many knots

The curve gets enough joints to chase every random wiggle in the sample — the same problem as “too wiggly” earlier.

The fix: a penalty against wiggleness. A “penalized” spline starts with plenty of knots, then a smoothness penalty pulls the curve back toward simplicity wherever the extra bend isn’t earning its keep. One “smoothing parameter” sets how hard that penalty pulls — the same balance the previous slide’s tuning setting was keeping.

REAL EXAMPLE · FLEXIBLE CURVE APPLIED TO SMOKING

The True Shape of Smoking Risk



--- Straight-line assumption

— What the flexible model actually found

The risk climbs steeply through the first ~40 pack-years — covering 69% of the smokers studied — then levels off. A straight-line model would have missed this bend entirely.

The bend is statistically confirmed, not noise ($p < 0.001$).

Author-created illustration based on Example 8, Figure 3 (N = 766 male smokers), not a reproduction of the original published figure; curve shape approximates the published pattern.

PUTTING IT TOGETHER

A Simple Decision Guide

Two questions — is this variable your actual interest, and does it have a “zero” spike? — point to the right approach:



It's just a nuisance to control for

Use fine stratification and conditioning so its effect cancels out mathematically.



It's the focus, no zero spike

Group it for a first look, then fit a flexible dose curve (fractional polynomials or a spline-based model).



It's the focus, with a zero spike

Group it with “zero” as its own baseline, then use the zero-aware version of those same flexible methods.

No method here is automatically wrong — the right pick depends on what question you are actually trying to answer. Two further cues: a nonmonotonic shape favors FP2 or an unconstrained spline, a monotonic one suits FP1; and zero-spike handling is well developed for fractional polynomials, but still more tentative for splines.

WRAPPING UP

Bottom Line

- 1 No single best method exists — the right choice depends on the goal and the data.
- 2 Grouping into categories is a reasonable first look, especially with a natural baseline like nonsmokers — but two categories alone is not recommended, and grouping shouldn't be the final word.
- 3 When the exposure itself is what you are studying, flexible modern methods beat assuming a straight line.
- 4 When a variable is just something to control for, the most careful matching available gives the most trustworthy results.
- 5 Reassuringly, careful analysts using different reasonable methods on the same good data usually reach very similar conclusions. Because the power to detect model misspecification is often low, though, similar-looking answers do not by themselves prove every model was adequate.

SOURCE & FURTHER READING

Becher, H. & Schmid, M. (2024). “Analysis of Continuous Covariates and Dose-Effect Analysis.”

In W. Ahrens & I. Pigeot (Eds.), Handbook of Epidemiology. Springer Science+Business Media.

This deck is a plain-language summary for a general audience. Anyone running their own analysis should consult the original chapter for the full statistical detail, formulas, and software code.

Prepared by Omokhodion Alfred Eriakha · September 19, 2026

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